Accounting for the High Density of Planet Mercury

Venus, Earth, and Mars are all thought to have formed from similar materials. Their iron cores take up similar fractions of the planets, and their silicate mantles- -the mantle is the layer of material between a planet's central core and its outer crust- -melt to produce similar kinds of volcanic rocks. By contrast, Mercury, the smallest and closest to the Sun of the four inner, or terrestrial, planets, may have been formed from a different bulk composition, or it may have formed under different conditions than the other terrestrial planets. The planet has a huge iron core, indicating either that the planet contains at least twice the iron content of the other terrestrial planets or that virtually all the iron from the bulk composition was pulled into its core, leaving its mantle almost iron-free. At the same time, the planet shows no recent volcanic activity, and so, unlike Venus, Earth, and Mars, its interior can be assumed to be stationary. The lack of volcanic activity is usually taken to mean that the planet has largely finished losing its internal heat. Mercury, however, has a small magnetic field. Magnetic fields are generally accepted to be caused by fluid movements in the mantle caused by heat loss. Somehow Mercury has created a balance that allows the generation of a magnetic field in the absence of mantle movements.

Mercury's average density is 341 pounds per cubic foot, just less than the densest planet, Earth, at 347 pounds per cubic foot. The planet's high density was first discovered by measuring the surprising acceleration of the Mariner mission toward Mercury in the grip of the planet's strangely strong gravity field. Earth is a much larger planet, and so its interior pressure is much higher, pressing the material into much denser forms. With the effects of pressure taken away, Mercury's density is actually much higher than Earth's, and it is therefore the densest planet in the solar system. The high density of the planet is thought to imply that it contains a lot of iron-more, by proportion, than any other planet. Lorn is the most common dense inner solar system element, and it is likely to be responsible for high density in Mercury. If Mercury's extra density is due to iron content, then Mercury has a core that is 60 percent or more by mass, the largest core relative to planetary radius of any planet in the solar system. Mercury's core is probably 1,120 to 1,180 miles in radius, nearly 75 percent of the planet's diameter. This is at least twice the metal content of Venus, Earth, or Mars.

Current models for planetary formation suggest that any terrestrial planet should form with a large percentage of silicates, which would make up a thick mantle above the core, as they do on Venus, Earth, and Mars. Mercury may have lost most of its silicate mantle in some catastrophic event after its early formation. The simplest candidate for this event is a giant impact. If a large body struck Mercury more or less directly, then its iron would combine with Mercury's into an unusually large core, and much of Mercury's silicate mantle could have been lost to space. The impacting body may also have been made predominantly of metal, like an iron meteorite. In this scenario, the metal meteorite would combine with the early, small pro to Mercury, creating a planet with an unusually large iron core. Alternatively, the silicate mantle could have been removed by vaporization in a burst of heat from the early Sun. A third possibility is that there may have been strong compositional variation in the early solar system and that in fact Mercury formed with the iron-rich composition it now seems to possess. Finally, Mercury could have differentiated in a particularly oxygen-poor environment. Without oxygen, iron will preferentially form a metal and sink into the core rather than forming iron oxide and combining into silicate mantle materials. In this scenario, Mercury could have started with the same bulk composition and ended with its current structure without having endured a giant impact.